**RESHEKESH B (192512123)**

**SIMULATION EXPERIMENT – 4**

# Design and Comparative Analysis of Butterworth, Chebyshev and Elliptic IIR Filters Using MATLAB MATLAB CODE

```
clc;
clear;
close all;

% Specifications
Fs = 1000;     % Sampling Frequency
Fp = 100;      % Passband Frequency
Fsb = 150;     % Stopband Frequency
Rp = 1;        % Passband Ripple (dB)
Rs = 40;       % Stopband Attenuation (dB)

% Normalize Frequencies
Wp = Fp/(Fs/2);
Ws = Fsb/(Fs/2);

%% Butterworth Filter
[Nb,Wb] = buttord(Wp,Ws,Rp,Rs);
[bb,ab] = butter(Nb,Wb);

%% Chebyshev Type-I Filter
[Nc,Wc] = cheb1ord(Wp,Ws,Rp,Rs);
[bc,ac] = cheby1(Nc,Rp,Wc);

%% Elliptic Filter
[Ne,We] = ellipord(Wp,Ws,Rp,Rs);
[be,ae] = ellip(Ne,Rp,Rs,We);

%% Display Filter Orders
```

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```
disp('Butterworth Filter Order');
disp(Nb);

disp('Chebyshev Type-I Filter Order');
disp(Nc);

disp('Elliptic Filter Order');
disp(Ne);

%% Display Filter Coefficients
disp('Butterworth Numerator');
disp(bb);
disp('Butterworth Denominator');
disp(ab);

disp('Chebyshev Numerator');
disp(bc);
disp('Chebyshev Denominator');
disp(ac);

disp('Elliptic Numerator');
disp(be);
disp('Elliptic Denominator');
disp(ae);

%% Frequency Response
[Hb,f] = freqz(bb,ab,1024,Fs);
[Hc,~] = freqz(bc,ac,1024,Fs);
[He,~] = freqz(be,ae,1024,Fs);

%% Magnitude Response Comparison
```

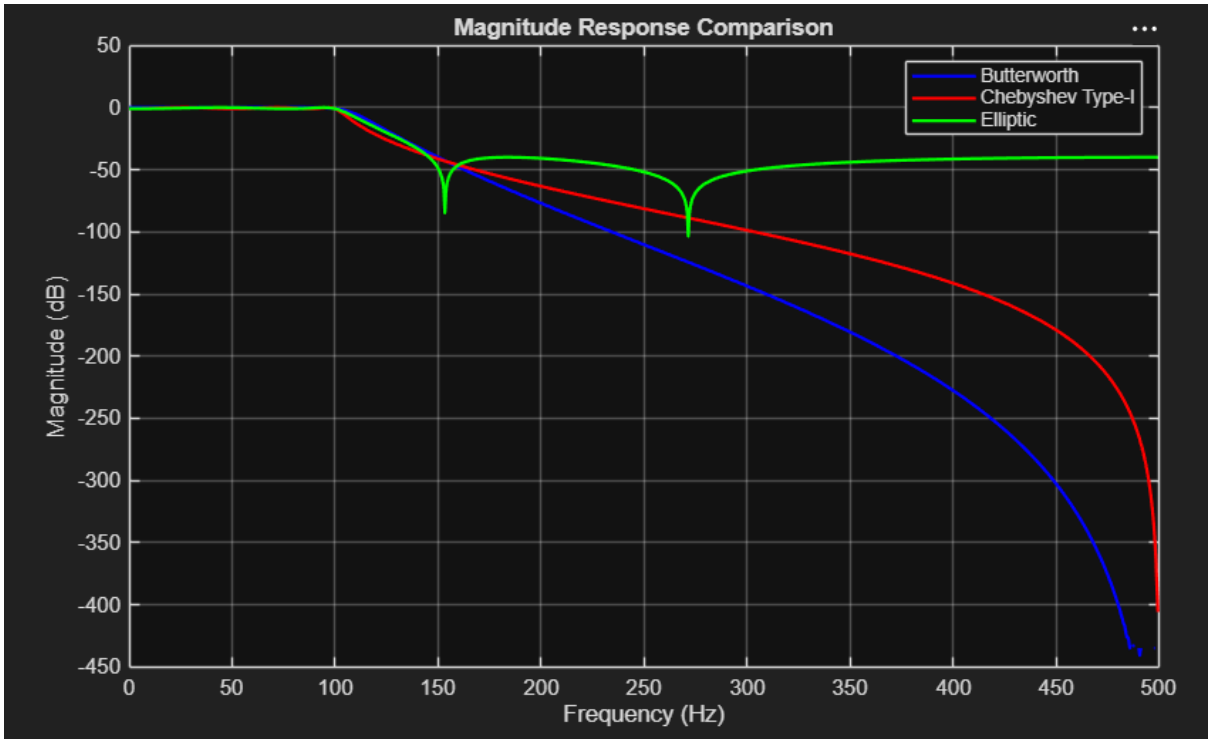
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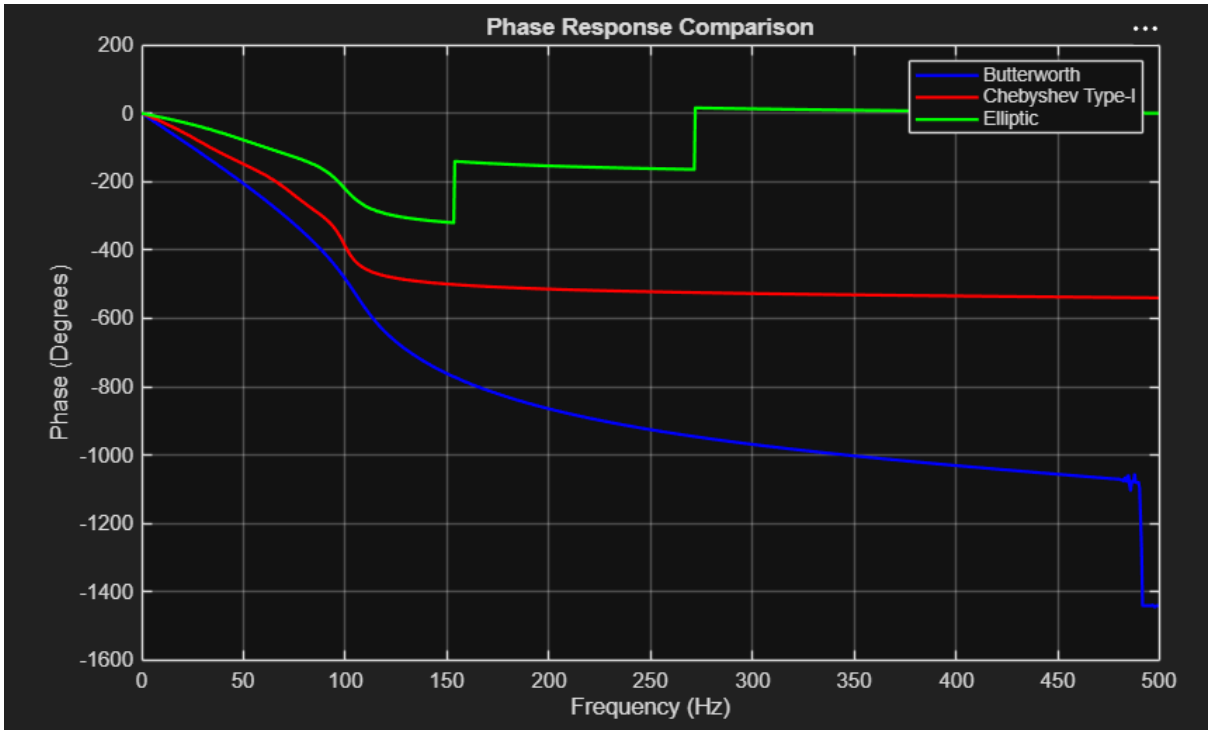
```
figure;
plot(f,20*log10(abs(Hb)),'b','LineWidth',1.5);
hold on;
plot(f,20*log10(abs(Hc)),'r','LineWidth',1.5);
plot(f,20*log10(abs(He)),'g','LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Magnitude Response Comparison');
legend('Butterworth','Chebyshev Type-I','Elliptic');

%% Phase Response Comparison
figure;
plot(f,unwrap(angle(Hb))*180/pi,'b','LineWidth',1.5);
hold on;
plot(f,unwrap(angle(Hc))*180/pi,'r','LineWidth',1.5);
plot(f,unwrap(angle(He))*180/pi,'g','LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');
title('Phase Response Comparison');
legend('Butterworth','Chebyshev Type-I','Elliptic');
```

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**OUTPUT**



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```
Butterworth Filter Order
    12

Chebyshev Type-I Filter Order
     6

Elliptic Filter Order
     4

Butterworth Numerator
   1.0e-03 *

    0.0002    0.0027    0.0146    0.0488    0.1098    0.1757    0.2050    0.1757    0.1098    0.0488    0.0146    0.0027    0.0002

Butterworth Denominator
    1.0000   -6.8844   22.4341  -45.5100   63.7725  -64.8501   48.9643  -27.6092   11.5218   -3.4662    0.7128   -0.0899    0.0053

Chebyshev Numerator
   1.0e-03 *

    0.0464    0.2782    0.6956    0.9274    0.6956    0.2782    0.0464

Chebyshev Denominator
    1.0000   -4.8694   10.3807  -12.3367    8.5969   -3.3268    0.5586

Elliptic Numerator
    0.0197   -0.0171    0.0333   -0.0171    0.0197

Elliptic Denominator
    1.0000   -3.0330    3.8118   -2.2911    0.5554
```

**Result:**

The Butterworth, Chebyshev Type-I, and Elliptic IIR filters were successfully designed using MATLAB for the given specifications. The filter coefficients, filter orders, and magnitude and phase responses were obtained and compared. It was observed that the Butterworth filter provides a smooth passband with the widest transition band, the Chebyshev Type-I filter offers a sharper transition with passband ripple, and the Elliptic filter achieves the sharpest transition with the lowest filter order by allowing ripple in both the passband and stopband.